
ECSE 4660/6660: Internetworking of Things

Lecture 16: Network Layer – Part IV:
NAT, ICMP, ICMPv6

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Spring 2026

Administrative

- Use Piazza for any questions
 - <https://piazza.com/rpi/spring2026/ecse46606660>
- Today's Quiz is available on **LMS**
 - Multiple choice questions
 - Due tonight 11:59 PM
- Lab
 - TA hours 4-5 PM daily in 5312 lab
 - Lab 3 due Mon, 3/9 (BLE, Zbee)
 - Lab 4 due Fri, 3/20 (DSR, AODV, ARP)
- HW
 - HW3 released, due Mon, 3/16
 - No Late submission allowed
- Remote class 2/26, ~~3/23~~ 3/30

- Project group formation
 - [IoT_SP26_project.xlsx](#)
- Project Next Steps
 - Work with TA to get the hardware needed (respond on piazza) – due TODAY 3/16
 - Mid-term report due 3/20 (<1 week)
 - In-class presentation+2min video 4/20, 4/23
 - Live-demo 4/27
 - Final code/report 5/1

Network layer: “data plane” roadmap

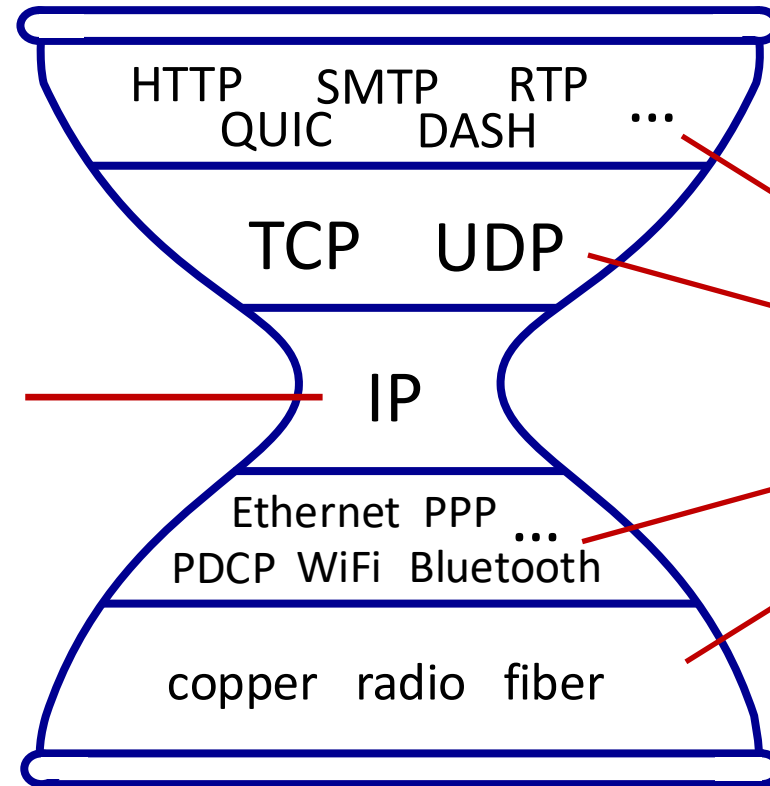
- Network layer: overview
- IP: the Internet Protocol
 - Datagram format and IP Addressing
 - DHCP
 - ARP
 - Routing Table Lookup
 - IPv6
 - NAT
 - ICMP
 - ICMPv6



The IP hourglass

Internet's "thin waist":

- *one* network layer protocol: IP
- *must* be implemented by every (billions) of Internet-connected devices

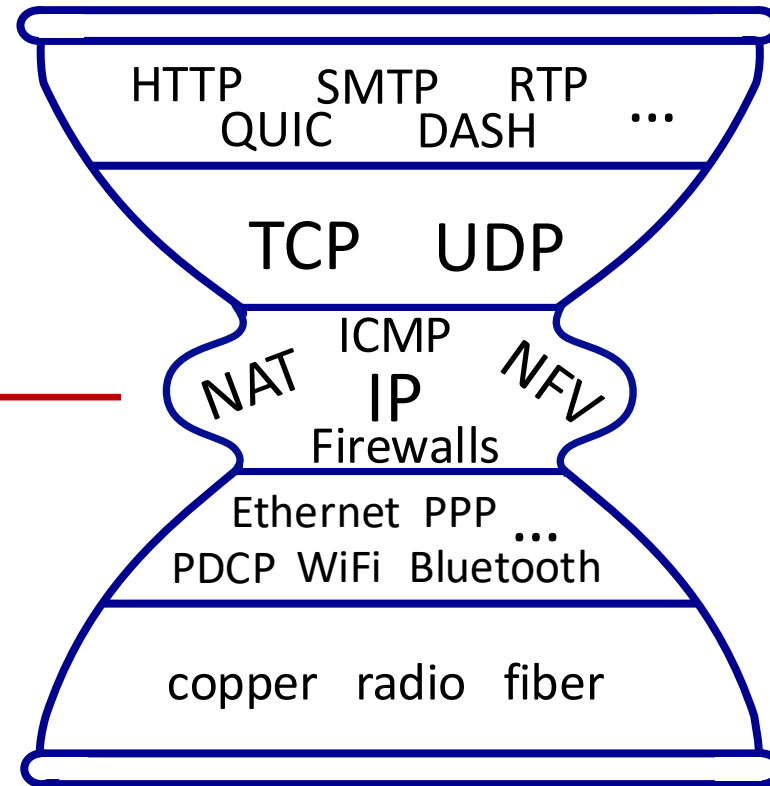


many protocols in physical, link, transport, and application layers

The IP hourglass, at middle age – Not only IP

Internet's middle age
"love handles"?

- middleboxes, — operating inside the network
- NAT, ICMP protocols

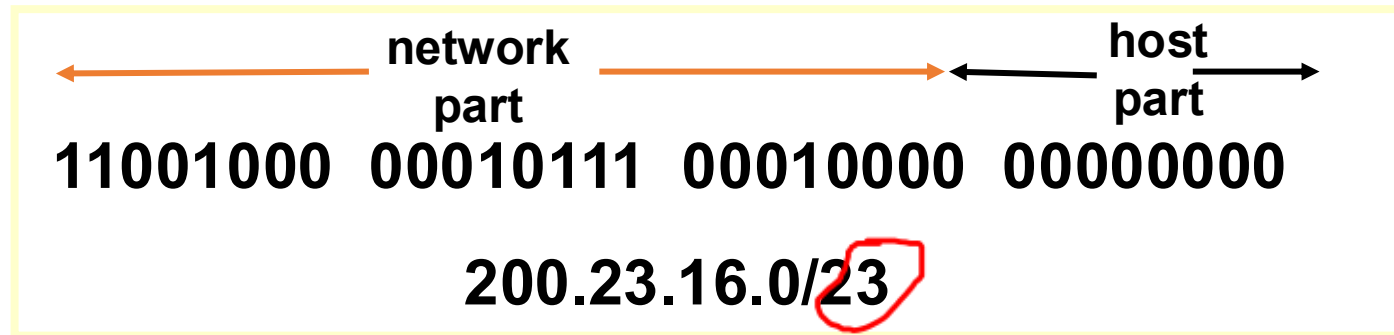


Classful vs. Classless Addressing - Recall

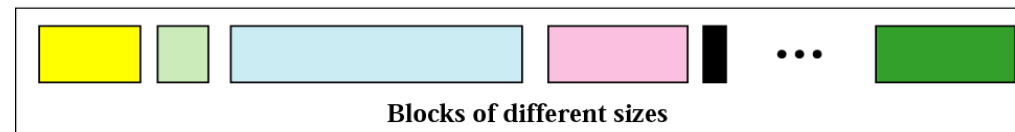
Limitation of classful addressing:

- inefficient use of address space, address space exhaustion (e.g., class B network allocated enough addresses for 65K hosts, even if only 2K hosts in that network)

- **CIDR: Classless InterDomain Routing**
network portion of address of arbitrary length
address format: a.b.c.d/x, x is # bits in network portion of address



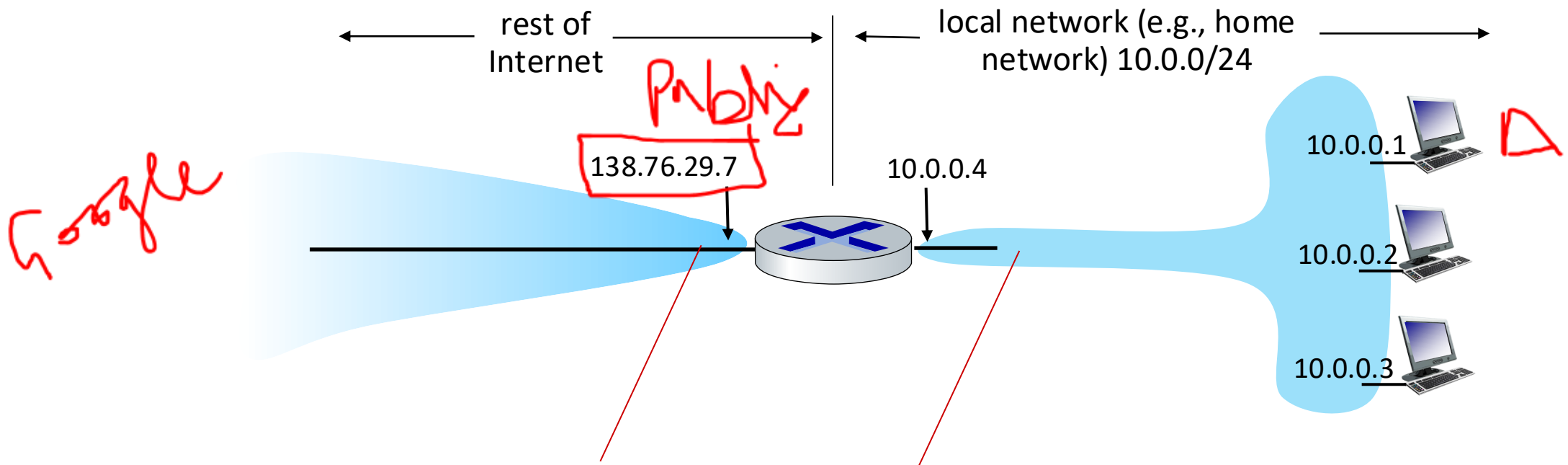
Address Space



[src: Forouzan]

NAT: network address translation

NAT: all devices in local network share just **one** IPv4 address as far as outside world is concerned



all datagrams *leaving* local network have *same* source NAT IP address: 138.76.29.7,
→ how to distinguish different devices?
use *different* source port numbers

datagrams with source or destination in this network have 10.0.0/24 address for source, destination (as usual)

NAT: network address translation

- all devices in local network have 32-bit addresses in a “private” IP address space (10/8, 172.16/12, 192.168/16 prefixes) that can only be used in local network
- advantages:
 - just **one** IP address needed from provider ISP for *all* devices
 - can change addresses of host in local network without notifying outside world
 - can change ISP without changing addresses of devices in local network
 - security: devices inside local net not directly addressable, visible by outside world

NAT: network address translation

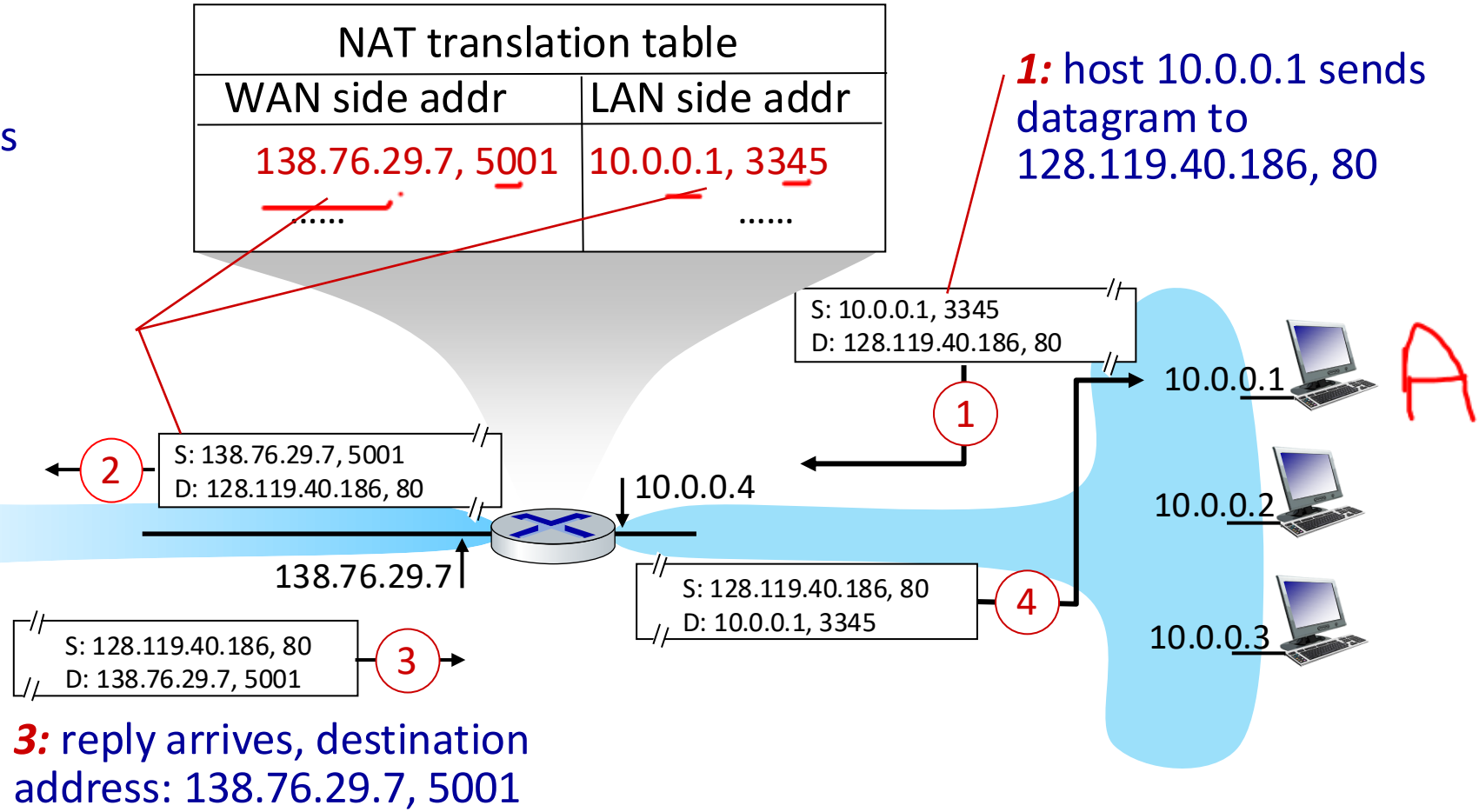
implementation: NAT router must (transparently):

- **outgoing datagrams:** **replace** (source IP address, port #) of every outgoing datagram to (NAT IP address, new port #)
 - remote clients/servers will respond using (NAT IP address, new port #) as destination address
- **remember (in NAT translation table)** every (source IP address, port #) to (NAT IP address, new port #) translation pair
- **incoming datagrams:** **replace** (NAT IP address, new port #) in destination fields of every incoming datagram with corresponding (source IP address, port #) stored in NAT table

NAT: network address translation

2: NAT router changes datagram source address from 10.0.0.1, 3345 to 138.76.29.7, 5001, updates table

1: host 10.0.0.1 sends datagram to 128.119.40.186, 80



NAT: network address translation

- NAT has been controversial:
 - routers “should” only process up to layer 3
 - address “shortage” should be solved by IPv6
 - violates end-to-end argument (port # manipulation by network-layer device)
 - NAT traversal: what if client wants to connect to server behind NAT?
- but NAT is here to stay:
 - extensively used in home and institutional nets, 4G/5G cellular nets

IPv6: Motivation – Alternate to NAT

- IPv4 limitations

- Shortage of address space
 - Subnetting, CIDR, dynamic resource allocation through DHCP, NAT only temporary solutions
- Need better support for real-time audio/video
 - No support for resource allocation/ reservation
- No encryption or authentication

- IPv6 addresses all the above IPv4 limitations

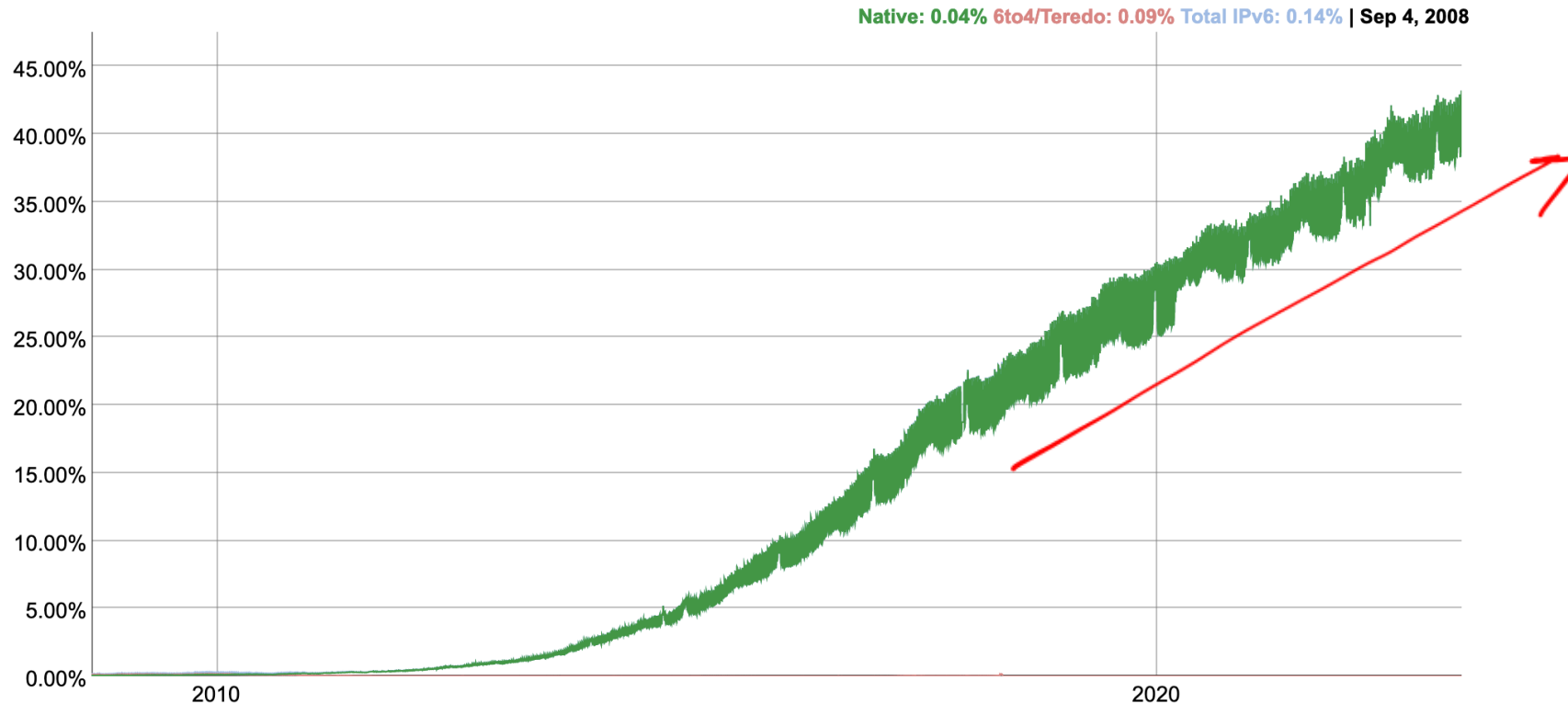
- Also has better header format and newer options

IPv6: adoption

- Google¹: ~ 40% of clients access services via IPv6 (2023)
- NIST: 1/3 of all US government domains are IPv6 capable

IPv6 Adoption

We are continuously measuring the availability of IPv6 connectivity among Google users. The graph shows the percentage of users that access Google over IPv6.



IPv6: adoption

- Google¹: ~ 40% of clients access services via IPv6 (2023)
- NIST: 1/3 of all US government domains are IPv6 capable
- Long (long!) time for deployment, use
 - 25 years and counting!
 - think of application-level changes in last 25 years: WWW, social media, streaming media, gaming, telepresence, ...
 - *Why?*

¹ <https://www.google.com/intl/en/ipv6/statistics.html>

Network layer: “data plane” roadmap

- Network layer: overview
- IP: the Internet Protocol
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 - DHCP
 - ARP
 - Routing Table Lookup
 - IPv6
 - NAT
 - **ICMP**
 - ICMPv6



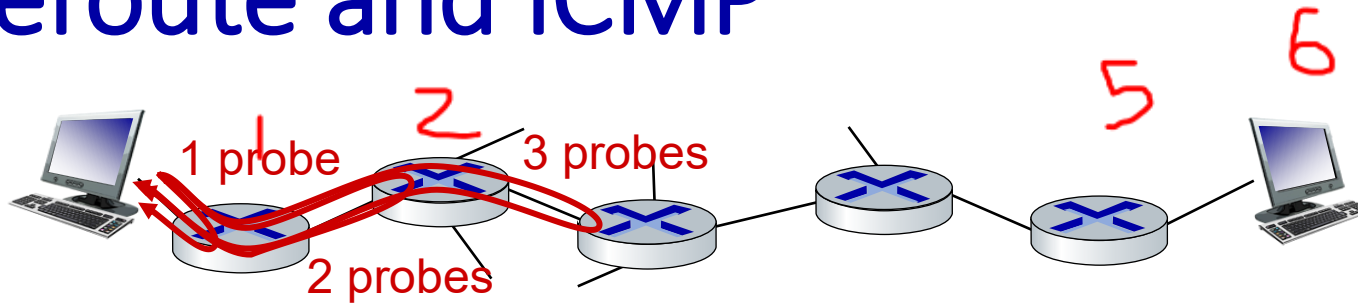
ICMP: internet control message protocol

- used by hosts and routers to communicate network-level information
 - error reporting: unreachable host, network, port, protocol
 - echo request/reply (used by ping)
- network-layer “above” IP:
 - ICMP messages carried in IP datagrams
- *ICMP message*: type, code plus first 8 bytes of IP datagram causing error



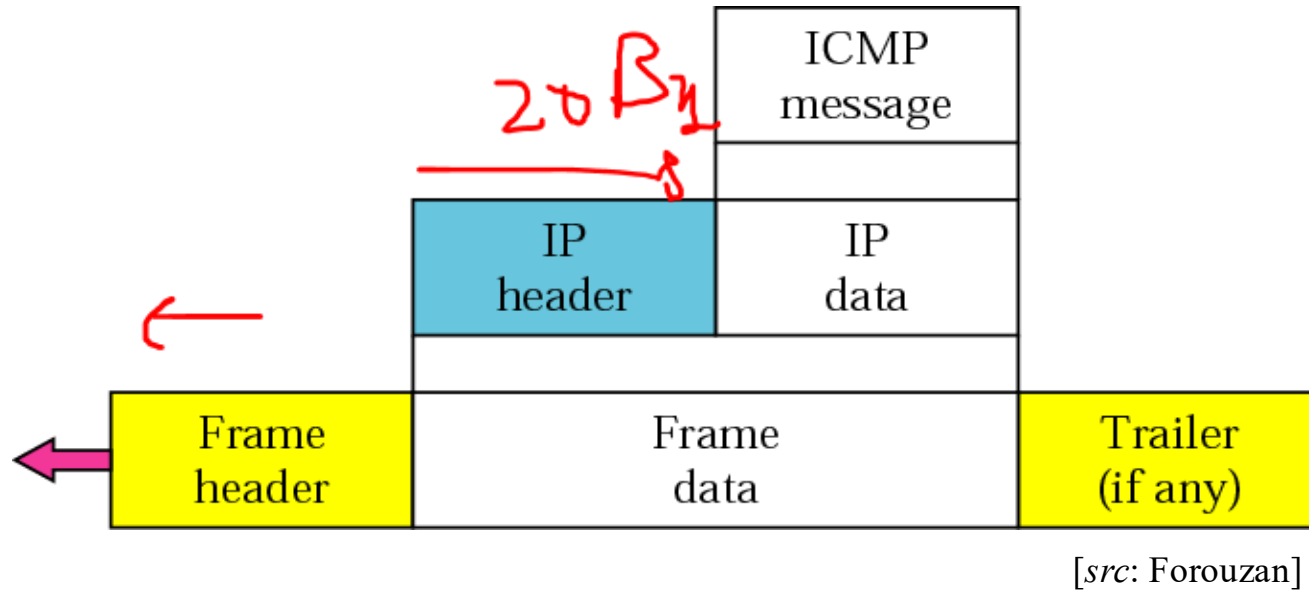
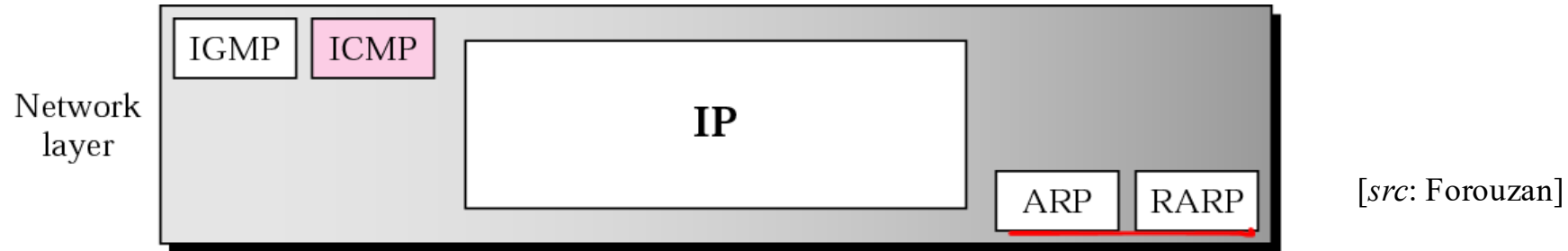
Type	Code	description
0	0	echo reply (ping)
3	0	dest. network unreachable
3	1	dest host unreachable
3	2	dest protocol unreachable
3	3	dest port unreachable
3	6	dest network unknown
3	7	dest host unknown
4	0	source quench (congestion control - not used)
8	0	echo request (ping)
9	0	route advertisement
10	0	router discovery
11	0	TTL expired
12	0	bad IP header

Traceroute and ICMP



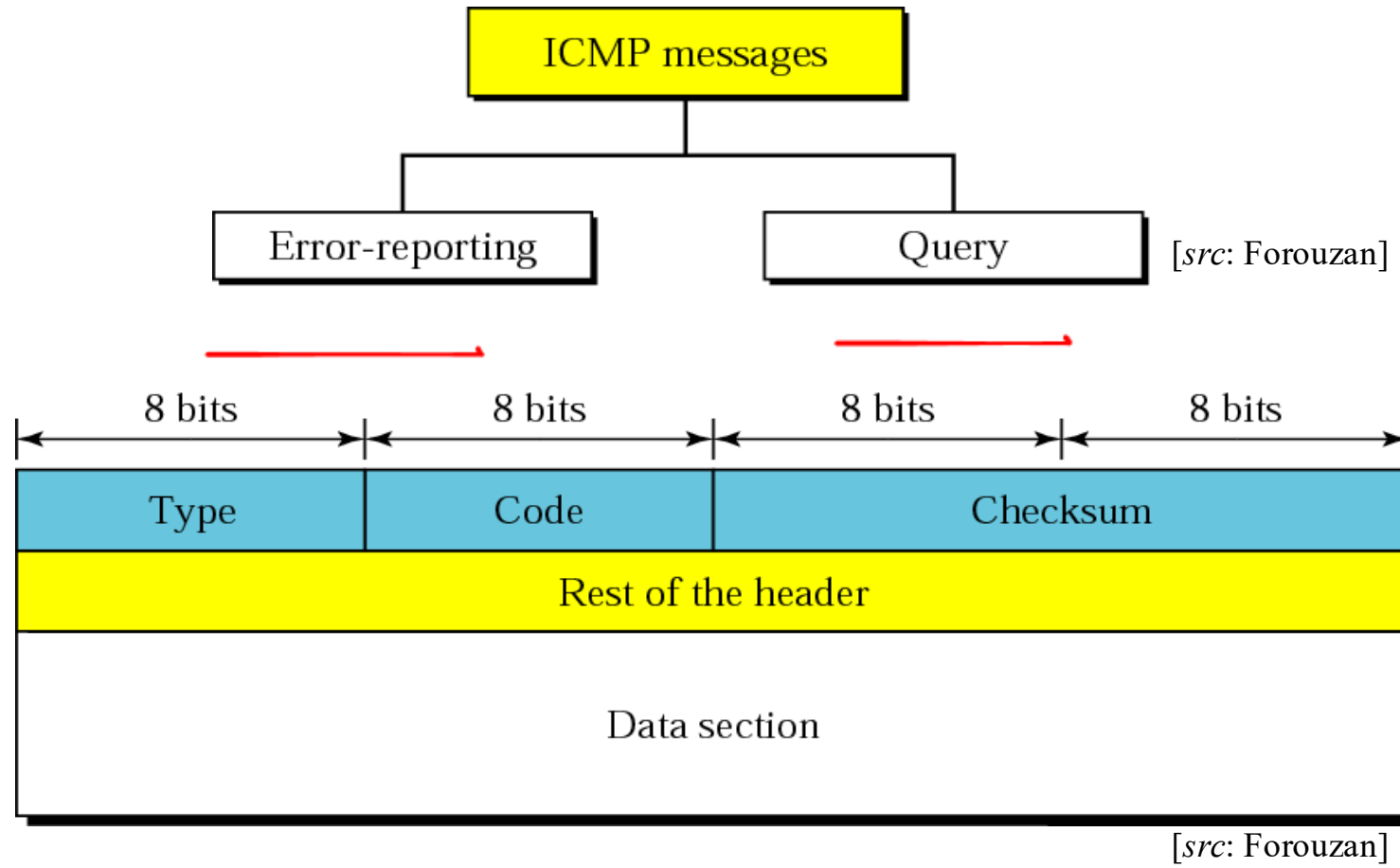
- source sends sets of UDP segments to destination
 - 1st set has TTL = 1, 2nd set has TTL=2, etc.
 - datagram in n th set arrives to n th router:
 - router discards datagram and sends source ICMP message (type 11, code 0)
 - ICMP message possibly includes name of router & IP address
 - when ICMP message arrives at source: record RTTs
- stopping criteria:
- UDP segment eventually arrives at destination host
 - destination returns ICMP “port unreachable” message (type 3, code 3)
 - source stops

ICMPv4 Format



ICMP Packet Encapsulation

ICMP Messages: Purpose & Format



ICMP Messages: Categories & Types

<i>Category</i>	<i>Type</i>	<i>Message</i>
✓ Error-reporting messages	3	Destination unreachable
	4	Source quench
	11	Time exceeded
	12	Parameter problem
	5	Redirection
✓ Query messages	8 or 0	Echo request or reply
	13 or 14	Timestamp request or reply
	17 or 18	Address mask request or reply
	10 or 9	Router solicitation or advertisement

[src: Forouzan]

ICMPv4 Error Reporting Messages

ICMPv4 Message Type	Code	Description
Destination Unreachable (Type 3)	0 - 15 (various codes)	Indicates that the destination is unreachable for various reasons such as network, host, port, or protocol unavailability.
Time Exceeded (Type 11)	0 – TTL Expired in Transit 1 – Fragment Reassembly Time Exceeded	TTL expired in transit occurs when a packet takes too long to reach the destination, often due to looping. Fragment reassembly time exceeded occurs when fragments are not reassembled within the allotted time.
Parameter Problem (Type 12)	0 – Pointer Indicates the Error 1 – Missing a Required Option	Indicates issues with incorrect parameters in the IP header that prevent packet processing.
Source Quench (Type 4)	0	Used historically for congestion control, but is now obsolete. It was sent to request the sender to slow down packet transmission.
Redirect Message (Type 5)	0 – Redirect for Network 1 – Redirect for Host 2 – Redirect for Type of Service and Network 3 – Redirect for Type of Service and Host	Used by routers to inform the sender about a better route for packet forwarding.

Example: Type 3 Destination unreachable

ICMP Type	ICMP Code	Meaning
3 (Destination Unreachable)	0	Network Unreachable
3	1	Host Unreachable
3	2	Protocol Unreachable
3	3	Port Unreachable
3	4	Fragmentation Needed and DF bit set
3	5	Source Route Failed
3	6	Destination Network Unknown
3	7	Destination Host Unknown
3	9	Communication with Destination Network Administratively Prohibited
3	10	Communication with Destination Host Administratively Prohibited

ICMPv4 Query Messages

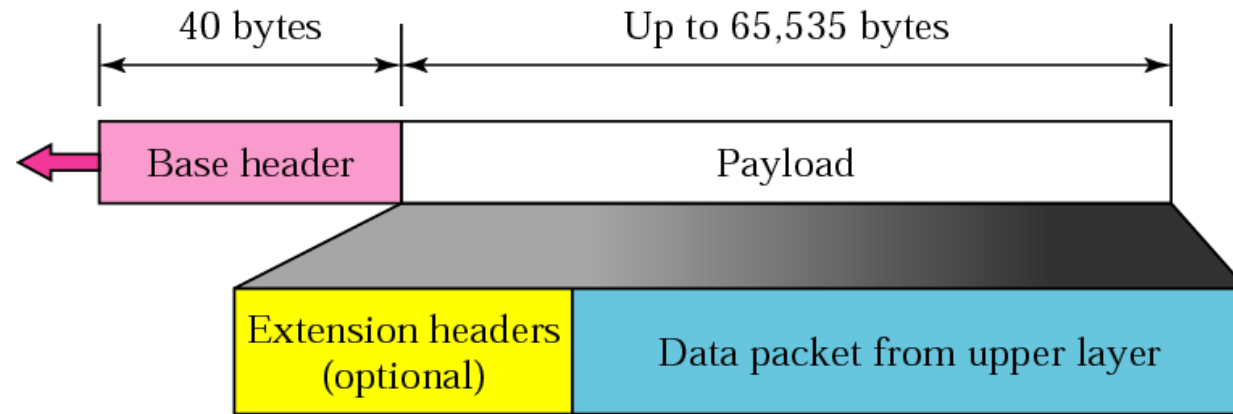
ICMPv4 Message Type	Code	Description
Echo Request / Reply (Type 8 / 0)	0	Used to check network connectivity and measure round-trip time. The Echo Request is sent, and the Echo Reply confirms reachability. Used in the ping command.
Timestamp Request / Reply (Type 13 / 14)	0	Used to synchronize time between devices. The Timestamp Request asks for a timestamp, and the Timestamp Reply returns it, allowing delay calculation.
Address Mask Request / Reply (Type 17 / 18)	0	Used to determine the subnet mask of a network. The Request is sent to a router, and the Reply provides the subnet mask. Mostly obsolete due to DHCP.
Router Solicitation / Advertisement (Type 10 / 9)	0	Used by hosts to discover routers. The Router Solicitation asks for routing information, and the Router Advertisement is sent by routers to announce their presence and provide routing details.

Network layer: “data plane” roadmap

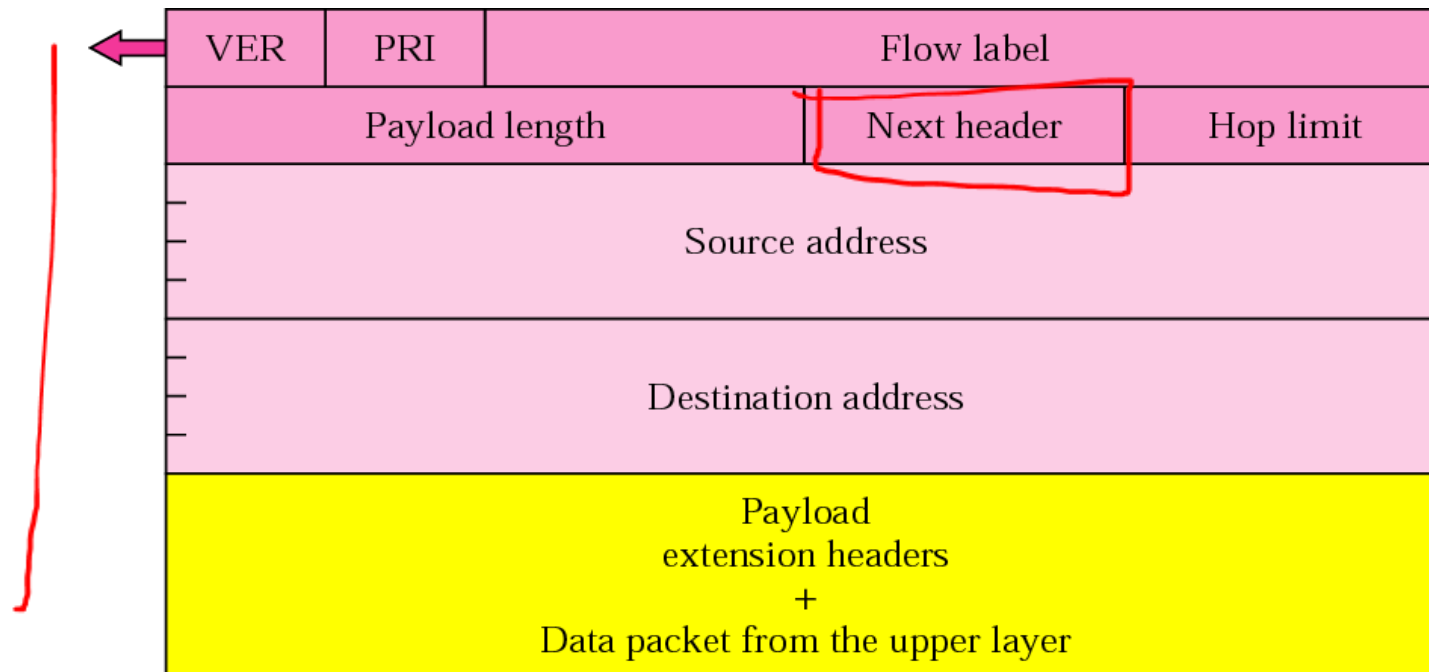
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Recap: IPv6 datagram format



[src: Forouzan]



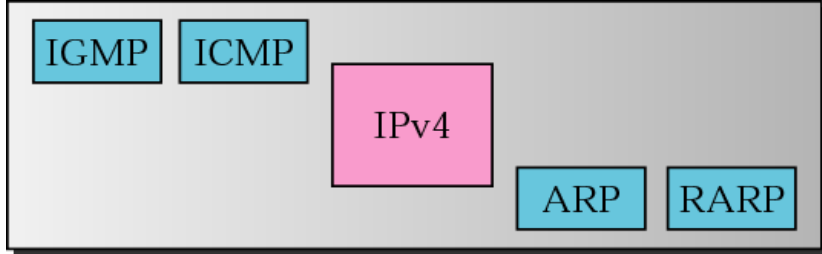
Next header codes

<i>Code</i>	<i>Next Header</i>
0	Hop-by-hop option
2	ICMP
6	TCP
17	UDP
43	Source routing
44	Fragmentation
50	Encrypted security payload
51	Authentication
59	Null (No next header)
60	Destination option

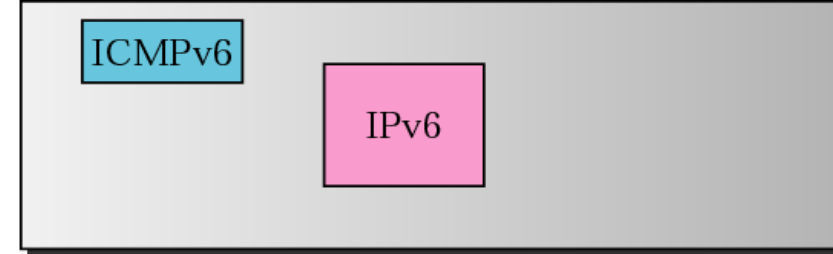
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IPv4 vs IPv6 network layers

- RARP dropped (BOOTP/DHCP used instead)
- ICMPv6 combines IGMP and ARP functionality



Network layer in version 4

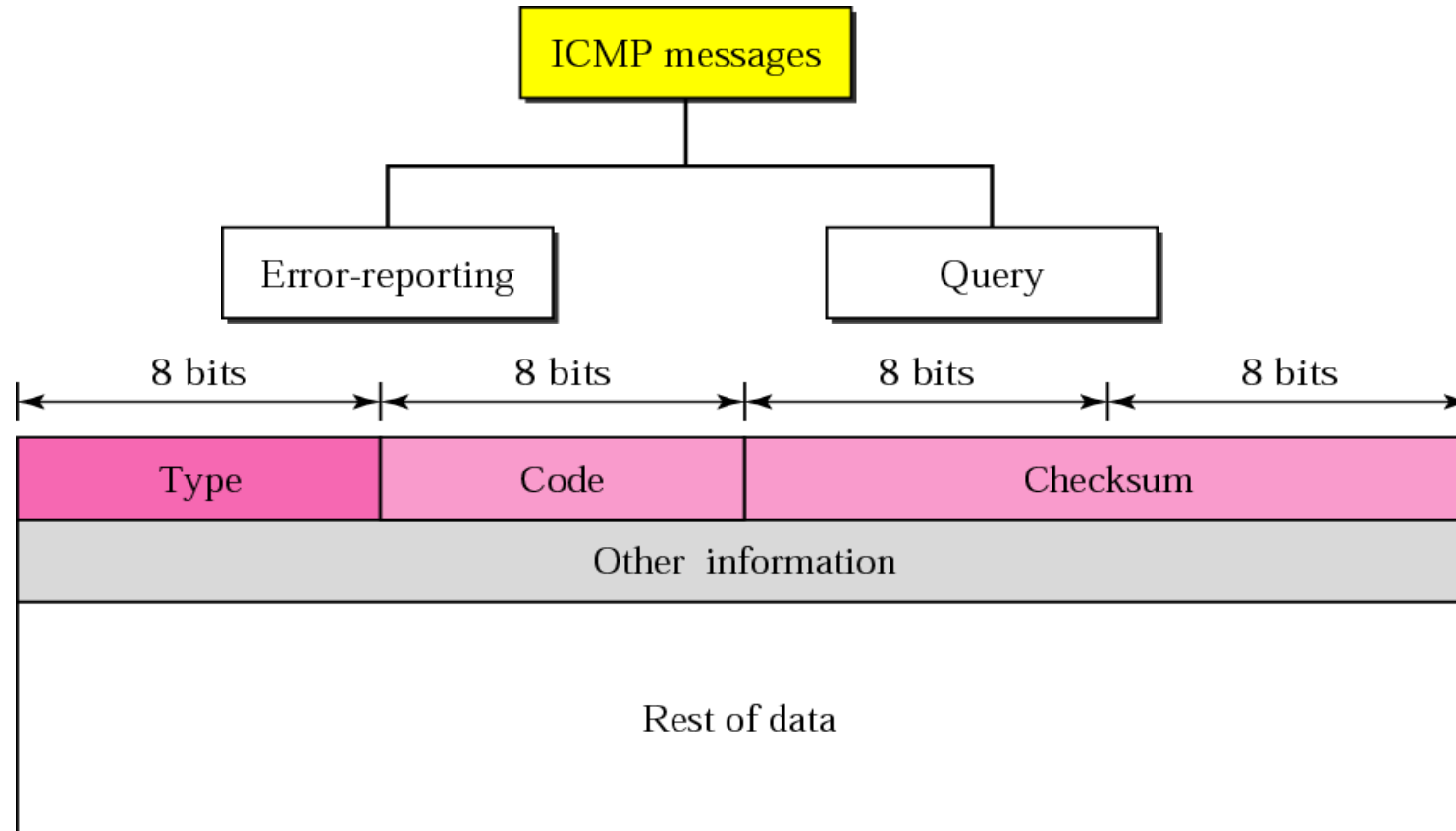


Network layer in version 6

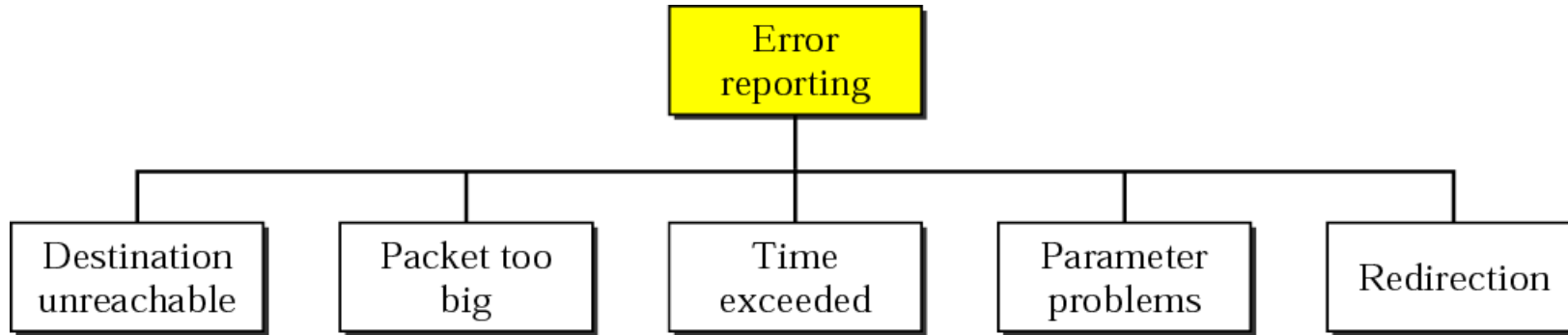
[src: Forouzan]

ICMPv6 message types and format

- Used for error/control reporting – IP does not have built in mechanism for error reporting
- Similar message types and format as in ICMPv4



ICMPv6 error reporting messages

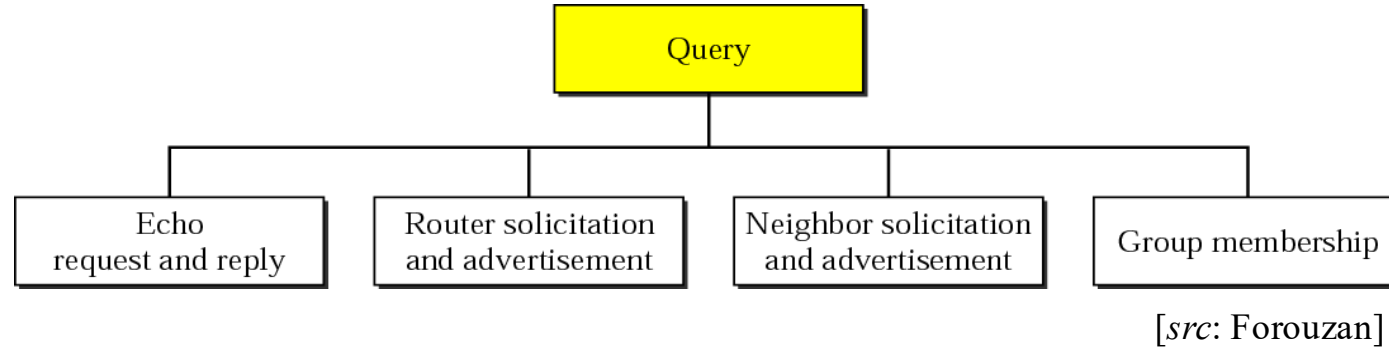


[src: Forouzan]

<i>Type of Message</i>	<i>Version 4</i>	<i>Version 6</i>
Destination unreachable	Yes	Yes
Source quench	Yes	No
Packet too big	No	Yes
Time exceeded	Yes	Yes
Parameter problem	Yes	Yes
Redirection	Yes	Yes

[src: Forouzan]

ICMPv6 informational/query messages



Type of Message	Version 4	Version 6
Echo request and reply	Yes	Yes
Timestamp request and reply	Yes	No
Address mask request and reply	Yes	No
Router solicitation and advertisement	Yes	Yes
Neighbor solicitation and advertisement	ARP	Yes
Group membership	IGMP	Yes

← It was rarely used, timestamp can be done by NTP

← Can be done through Router Advertisement

[src: Forouzan]

Router solicitation and advertisement messages

- Solicitation message: host's physical address can be optionally included to help the router respond
- Router advertisement message:
 - The router announces only itself (not other routers it knows of)
 - Router's physical address, MTU can optionally be included



Neighbor solicitation and advert messages

- Neighbor solicitation: Sender's physical address can be optionally included, to help the router respond
- Neighbor advertisement: Target's physical address included as an option

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- Routing Protocol for Internet
- Routing Protocol for IoT

